# Escape speed

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain escape in an ideal gravitational-energy model.

**SUCCESS CRITERIA - I can...**

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “escape speed” and “orbit” to explain your model or result. Sentence starters are optional.

**escape speed**: minimum ideal speed to leave without further thrust

**orbit**: a path around another body under gravity

**radius**: distance from the center to the edge of a circle or sphere

## 01 Warm-up: bring back an earlier idea

Total kinetic energy changes from 15 J to 12 J in an isolated collision. Is it elastic? Where could the difference go?

**Word help:** kinetic energy = energy associated with motion

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

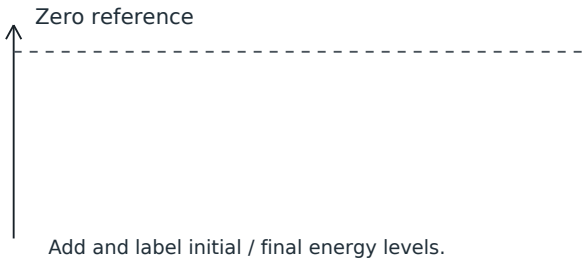
## 02 Worked example: explain the model

**Reference:** $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

**Worked reasoning:** For a simulated planet with $GM=50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)}=5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

**Your turn: Draw** initial kinetic and gravitational-potential-energy bars whose total is zero.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




Add and label initial / final energy levels.

Label K/m, U/m and total E/m in J/kg.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

Does escaping mean that gravity turns off at the launch point?
**Explain** what the speed allows in an energy model.

**Sentence starter:** My claim is … . The diagram/data show …, so … .

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 04 Apply the physics

A model needs 50 J to reach infinity. A projectile begins with 60 J kinetic energy. What energy is left at infinity, neglecting losses?
**Use** the energy model to show the quantities you compare; label units and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** My design shows … . It meets the requirement by … .

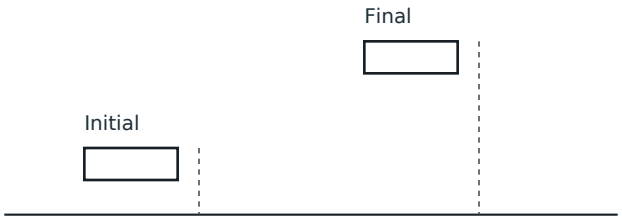
**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

Lift a 6 kg object by 8 m.
**Use** g=10 N/kg.
**Find** the change in gravitational potential energy.
**Label** the distances or energy levels used in your reasoning.

**Word help:** potential energy = energy associated with the configuration of interacting objects

**Sentence starter:** I found … [unit]. This value means … in this situation.




Choose and label a zero-height level.

**Marzano check - circle:** 1 2 3

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Continue down each column.

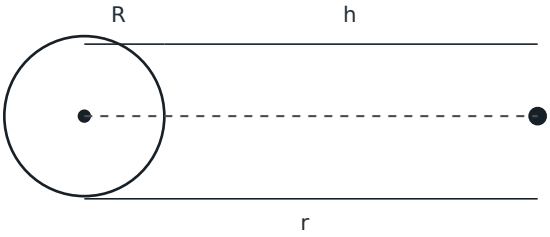
Sentence starters are optional. You may plan in any language.

## 07 Exit check

An object has less energy than required to escape and receives no further thrust. Can this ideal model expect it to escape?
**Use** the energy model to show the quantities you compare; label units and any unknowns.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** Both … . They differ in … because … .

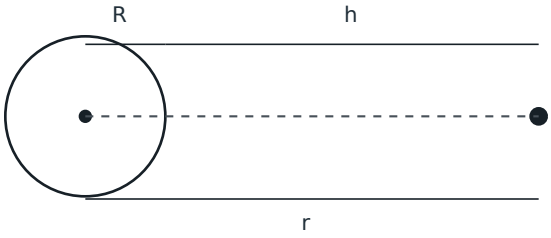



**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

The same simulated planet's circular orbital speed at its surface is supplied as 3.54 m/s.
**Compare** it with escape speed 5 m/s and explain their different purposes.
**Use** the energy model to show the quantities you compare; label units and any unknowns.

**Sentence starter:** Both … . They differ in … because … .




**Marzano check - circle:** 1 2 3